Naman Agrawal

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Research Interests

Deep Learning • Algorithm Design • Natural Language Processing • Bayesian Methods • Computational Social Science • Game Theory • Applied Econometrics

Education

National University of Singapore *Aug 2021 – Jul 2025*
Bachelor of Science (Honours) in Data Science and Analytics; GPA: 4.94/5.00 Singapore

Minor in Computer Science; Specialization in Statistical Methodology
Award: Lijen Industrial Development Medal for Best Honours Thesis
Coursework: Design & Analysis of Algorithms, Large Language Models, Databases, Media Computing, Bayesian Statistics, Numerical Computation, Convex Optimisation, Real Analysis, Multivariable Calculus, Linear Algebra

National University of Singapore *Aug 2021 – Jul 2025*
Bachelor of Social Science (Honours) in Economics; GPA: 4.98/5.00 Singapore

Specialization in Monetary & Financial Economics
Award: Paul Sherwood Memorial Gold Medal for Best Graduate
Coursework: Advanced Micro & Macro Economics, Econometrics, Game Theory & Mechanism Design, Financial Economics, Money & Banking, International Finance, Behavioural Economics, Applied Time Series Analysis

Work Experience

Department of Economics, National University of Singapore *Aug 2025 – Present*
Predoctoral Researcher under Dr. Ruben Durante and Dr. Xiaoyue Shan Singapore

- Built large-scale data pipelines for web scraping, entity resolution, geospatial matching, NER, and metadata extraction across diverse datasets, including 200k+ news articles, the universe of Freelancer projects, and longitudinal street-name data covering all Spanish municipalities. Used these datasets to study conflict coverage, labor market behavior, symbolic politics, and the evolution of online work in the wake of gen AI.
- Developed computational methods to study political behavior, combining deep learning, organizational network data, and structured databases. Projects included LLM-based detection of political dog whistles across election cycles, predictive modeling based on peer characteristics for supervisor-student matching, and identifcation of potential conflicts of interest by linking media coverage to corporate ownership structures in Orbis.
- Designed computational infrastructure for large-scale field experiments including 20+ experimental task websites, survey instruments, and automated application systems. Deployed these tools on BDJobs and Upwork to study gender, expectations, political preferences, and labor-supply decisions.

Procter & Gamble and Institute of Mathematical Sciences *May 2024 – Jul 2024*
Research Intern in Statistics (Bayesian Dynamic Borrowing) Singapore

- Collaborated with P&G's research team to develop adaptive trial-design algorithms that leverage historical data to reduce control-group sample sizes in randomized controlled trials while preserving statistical validity and Type I/II error control. Applied these methods to oral care and face cream trials, evaluating their performance across varying degrees of congruence between historical and current trial data.
- Optimized Bayesian borrowing techniques, including normalized power priors, commensurate priors, elastic priors, Bayesian hierarchical models, and meta-analytic-predictive priors, to improve borrowing accuracy in trials with low historical congruence, while reducing computational overhead by up to 78%.

School of Computing, National University of Singapore *Nov 2022 – Feb 2023; May 2024 – Aug 2024*
Research Intern in Financial Data Science Singapore

- Examined firm alliance networks and their temporal effects on profitability, utilizing Louvain community detection and network centrality measures to understand how dynamic inter-firm networks impact long-term performance.
- Explored several ensemble techniques and base learners for XGBoost models to optimize financial fraud prediction.

Singapore Bus Services Transit *Jul 2023 – Mar 2024*
Research Intern in Machine Learning (Computer Vision) Singapore

- Developed deep convolutional models to detect visual impairments in surveillance footage, including blur and overexposure, achieving 96% classification accuracy across heterogeneous scenes ranging from high-contrast stationary backgrounds to roads with continuous vehicle activity.
- Designed a custom Siamese architecture combining SVD-based and Fourier-domain representations for robust blur detection, distinguishing genuine image degradation from motion-induced artifacts across diverse conditions.

Jio AI Centre for Excellence *May 2023 – Aug 2023*
Research Intern in Machine Learning (3D Computer Vision) India

- Researched algorithms for Simultaneous Localization and Mapping (SLAM): attention-based graph feature matching (SuperGlue), 3D reconstruction via Neural Radiance Fields (NeRF), and monocular depth estimation (ZoeDepth).
- Experimented with alternative training methods, including Forward-Forward training and Predictive Coding, optimizing code for 3x performance improvement and enhanced scalability through efficient gradient computation.

A.P. Moller - Maersk *May 2022 – Aug 2022*
Data Analytics and IT Intern Singapore

- Developed an Electronic Virtual Management System (EVMS) for APM Terminals, streamlining logistics, finances, operations, and risk management by collaborating with stakeholders to understand and address their requirements.
- Integrated 20+ databases enabling automated report generation, real-time dashboarding for enhanced visibility, and systematic tracking of operational KPIs.

Conferences and Publications

Agrawal, N. (2026). Phase-Aware Deep Learning with Complex-Valued CNNs for Audio Signal Applications. *IEEE Conference on Artificial Intelligence (CAI) 2026*, Granada, Spain.
Proposed complex-valued CNN architectures and training adaptations (complex batch-norm, weight initialization, complex activations) to exploit phase information ignored by real-valued nets. Evaluated in three stages: image baselines, MFCC audio classification, and phase-aware GNNs. Results show that complex architectures can match real CNNs while offering improved handling of phase in audio pipelines & highlight trade-offs in stability versus expressivity across different complex activations.

Agrawal, N. Crisis, Compassion, and Capital: The Economics of Global Charitable Giving. *Presented at the Singapore Economics Review Conference (SERC) 2026*, Singapore.
Used the Russo-Ukraine war as a natural experiment and analyzed nearly 50,000 GlobalGiving projects across 201 countries to study whether major crises create new charitable capital or reallocate existing donor attention. DiD estimates show that 50-70% of the surge in Ukraine-related giving reflects genuine additionality from new donors, while 30% comes from substitution away from other causes, with emotionally less salient projects experiencing the largest crowding-out effects.

Agrawal, N., Shariffdeen, R., Wang, G., Rasnayaka, S., & Iyer, G. N. (2025). Analysis of Student-LLM Interaction in a Software Engineering Project. *IEEE/ACM International Workshop on Large Language Models for Code (LLM4Code), ICSE 2025*, Ottawa, Canada.
Analyzed 13 weeks of student–LLM interaction, 730+ generated code snippets, and human intervention patterns to characterize how conversational interaction affects code generation quality and complexity. Results showed that interactive LLM use produced lower-complexity and higher-quality code than automated generation, identifying interaction design as an important determinant of LLM-assisted programming performance.

Software

Agrawal, N. (2025). `covalchemy`: R Package to Construct Joint Distributions with Control of Statistical Properties.
Provides tools for constructing multivariate distributions that preserve empirical marginals while allowing precise control over correlation structure, mutual information, and higher-order dependence. Benchmark experiments demonstrate that the methods maintain marginal fidelity with less than 1% error while enabling precise manipulation of dependence structures for research applications and statistical pedagogy.

Agrawal, N., Gopal, V. et al. (2026). `autoharp`: R Package for Semi-Automatic Grading of R/Rmd Scripts.
Provides a customizable toolkit for assessing and grading student R and R Markdown submissions, including automated correctness checks, run-time statistics, and static code analysis. Supports workflow automation for educators with functions to render and evaluate scripts, extract code structure, and generate comprehensive diagnostic reports.

Working Papers

Agrawal, N. MOSAIC: Set-Based Multi-Bit Watermarking for Language Models.
Developed a novel set-based multi-bit watermarking framework that recovers embedded payloads from unordered token parity constraints using belief propagation. Demonstrated over 95% exact payload recovery on Llama-2-7B for up to 32-bit messages while remaining robust to token reordering, insertion, and deletion without relying on model-specific decoding.

Agrawal, N. Equilibrium Analysis for Strategic Information Revelation in Data Markets.
Developed a tractable signalling model where data sellers choose how much of a dataset to disclose and buyers update beliefs accordingly. Derived conditions for pooling versus separating equilibria and used simulations to demonstrate how disclosure costs and prior beliefs shape market vulnerability to adverse selection.

Agrawal, N. Crowding In or Crowding Out? Donor-Level Evidence on Private Charity and the Retreating State.
Used the 2025 U.S. federal government shutdown as a natural experiment and analyzed 32.6 million GoFundMe donations to study how private charity responds to government withdrawal. Showed that new donor entry more than offsets individual-

level substitution, but overall replacement remains economically negligible and highly unequal, with funding concentrated in wealthier states and campaign success depending more on deservingness narratives than on described material need.

Agrawal, N. The Revealed Value of a Rare Bird: Travel-Cost Valuation of Biodiversity from Vagrant Bird Sightings. Developed a novel travel-cost framework using exogenous vagrant bird appearances and 13,271 eBird visits across India to estimate the recreational value of biodiversity. Identified the causal effect of rare species on birdwatching demand, recovering a revealed consumer surplus of approximately $586 per rare-bird event. Distinguished participation from travel-distance responses and showed that rarity and novelty, rather than aesthetic appeal, drive recreational value.

Agrawal, N. Rigidity as a Signal: A Game-Theoretic Explanation for Vacancies in Urban Real Estate Markets. Modeled vacancy as a deliberate, costly signal in a two-period game with heterogeneous sellers, deriving multiple equilibrium types including pooling-holdout, pooling-discount, and separating forms. Analyzed comparative statics and demonstrated how targeted policy levers such as vacancy taxes and price-ratio mandates can reshape equilibria towards more informative outcomes and improve allocative efficiency in urban housing markets.

Work in Progress

Agrawal, N. Cultural Flows in the Digital Age: Evidence from YouTube's Global Music Charts.

Agrawal, N. The Evolution of the Supreme Court of India: Judicial Decision-Making, Citation Networks, and Institutional Change.

Professional Service

Referee: IEEE Conference on Artificial Intelligence (IEEE CAI) 2026.

Teaching Experience

Department of Computer Science & Economics, NUS *Jul 2022 – Jul 2025*
Undergraduate Teaching Assistant Singapore

Design and Analysis of Algorithms (CS3230), Data Structures and Algorithms (CS2040), Artificial Intelligence: Technology and Society (IT1244), Programming Methodology (CS1010S), Computational Media Literacy (NM2207), Data Visualization (DSA2101), Introduction to Data Science (DSA1101), Macroeconomics (EC2102).

Honors & Awards

Paul Sherwood Memorial Gold Medal (Best Graduate in Economics)	2025
Lijen Industrial Development Medal (Best Honours Project)	2025
Sugar Industry of Singapore Prize (Best Performance in Faculty of Science)	2022
NUS Dean's List & Dean's Scholars List	2022–25
Undergraduate Teaching Excellence Award, Department of Economics	2025
Undergraduate Teaching Excellence Award, Department of Economics	2024
School of Computing Honour List of Tutors	2025
School of Computing Honour List of Tutors	2024
Department of Statistics & Data Science Teaching Assistant Award	2025
Department of Statistics & Data Science Teaching Assistant Award	2024
Top Student Awards: Data Structures & Algorithms, AI, Database Systems	2023–25
National Economics Olympiad (SRCC): All India Rank 2	2020

Selected Activities

Volunteering: Autism Resource Centre (Google Data Analytics Mentor, 2022), NUS Community & Social Service Club (Art Volunteer for Project HeART, 2022), Sembawang Family Service Centre (Python Tutor, 2022), Centre for Computing for Social Good (AI Primer Tutor, 2022), Anmol Society (English Teacher, 2019-2020).

Leadership: PGP Residence (Cluster Leader), NUS Data Science Society (Research Director), NUS Math Society (LaTeX, Director), NUS Economics Case Comp (Research Executive), NUS Economics Camp (Publicity Executive).

Writing: Published 25+ articles with 100,000+ reads in Towards Data Science and Analytics Vidhya on probability theory, deep learning, and statistical analysis.

Competitions: Advent of Code (2023–24), NUS Hack & Roll (2024), IMC Prosperity Algorithmic Trading (2024), SMU Future Economic Forum (2023–24), Citadel APAC Datathon (2022).

Workshops Organized: Generative AI for Business, Computer Vision with OpenCV, Network Analysis with tidygraph (R User Conference), Introductory & Advanced LaTeX, NUS SoC Summer Workshops.

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